

The empirical recurrence for OEIS A241116, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A241116 records “Empirical recurrence of order 74” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 83$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 76$. Its last coefficient is $c_{74} = -42845606719488 \neq 0$, so no further tail satisfies anything shorter; any order-74 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A241116 is “Number of $n \times 4$ 0..3 arrays with no element equal to exactly three horizontal or vertical neighbors, with new values 0..3 introduced in row major order.”. It has offset 1 and begins

15, 2614, 587555, 134229632, 30679522712, 7012241396116, 1602748583683541, 366331237082570675, ...

The entry states:

Empirical recurrence of order 74 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 11:10:31, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 83$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 83$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 74: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 74.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 166$ exact terms from $n = 2$ on, it returns order 74 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{74} c_i a(n-i),$$

c_1	226	c_{20}	−1246467103020400	c_{39}	−5106452587324796218	c_{58}	−2699410839618021
c_2	693	c_{21}	−5508255352371160	c_{40}	2467613681967678752	c_{59}	−460730960111588
c_3	−23678	c_{22}	25047984761949083	c_{41}	17039563531521605638	c_{60}	3951256738151363
c_4	−181091	c_{23}	−921368268294190	c_{42}	10781217014628719771	c_{61}	3032875721813770
c_5	2213828	c_{24}	−59730699027083090	c_{43}	69382524442948718	c_{62}	2623275880958200
c_6	5325009	c_{25}	40631378521459194	c_{44}	12156151484816540654	c_{63}	1253123299255396
c_7	−276870298	c_{26}	−13979110913298979	c_{45}	5095771529010018944	c_{64}	−133660449588843
c_8	1185266447	c_{27}	−45403085572819750	c_{46}	−14462049206808821399	c_{65}	−119836842379154
c_9	14468765092	c_{28}	384123995073927622	c_{47}	−9966617809907836848	c_{66}	−150392086112016
c_{10}	−93388379888	c_{29}	−430823724265766858	c_{48}	−13000815465243800430	c_{67}	−100629297614368
c_{11}	−253358538544	c_{30}	−1555140420011068517	c_{49}	−12543740357289226904	c_{68}	172874883437133
c_{12}	2599824671183	c_{31}	521706890146161660	c_{50}	−3683840406530978372	c_{69}	176720601388921
c_{13}	2209740578898	c_{32}	2946226576031067318	c_{51}	7929510056180968806	c_{70}	−3787193117952
c_{14}	−39166542398674	c_{33}	2215767418747375092	c_{52}	1690994771617906494	c_{71}	−19028772024410
c_{15}	2200557632876	c_{34}	1805361639283735183	c_{53}	10623207271429383902	c_{72}	−27155386851379
c_{16}	314837262211337	c_{35}	−1735976654403425770	c_{54}	6230748871483056159	c_{73}	−16305133668249
c_{17}	−228140561587880	c_{36}	−8179939007786373917	c_{55}	685473548137075674	c_{74}	−4284560671948
c_{18}	−1216926903837371	c_{37}	−6318742757162596574	c_{56}	−1585439530418791908		
c_{19}	2057330996717578	c_{38}	−236870957465717171	c_{57}	−1397536883069218002		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 76$, and it is the recurrence OEIS A241116 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{74} - c_1 x^{74-1} - \dots - c_{74}$. Its constant term is $-c_{74} = 42845606719488$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 74 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 74, so $\deg q = 74$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 74, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 74, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -42845606719488 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-74 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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